Faculty of Information Systems
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Editor-in-Chief
Informatics in Medicine Unlocked

Dear Editor-in-Chief,

Please consider our manuscript, "Architecture and Preprocessing Choices under Dataset Shift in Single-Channel EEG Sleep Staging: A Controlled Evaluation on Sleep-EDF and SHHS1," for publication as an Original Research article in *Informatics in Medicine Unlocked*.

Automatic sleep staging is a medical-data analysis problem in which performance can be sensitive to the cohort, acquisition system, EEG derivation and preprocessing pipeline. Our study compares six active configurations on shared subject-wise Sleep-EDF partitions and evaluates locked transfer, without weight updates, on 180 SHHS1 participants. It combines subject-level inference with computational measurements and class-specific error analysis. The main finding is deliberately bounded: the evaluated ResNet-1D–TCN pipeline was faster and improved aggregate cross-cohort performance relative to the CNN–BiLSTM control, but both model families retained substantial N3 under-detection. Preprocessing also produced a material cross-cohort contrast under a fixed architecture and training recipe.

We believe the manuscript fits the journal because it examines the reliability and practical evaluation of automated EEG analysis rather than presenting an architecture-only benchmark. This positioning is consistent with prior IMU work on automatic sleep scoring using recurrent, convolutional and knowledge-guided methods. The manuscript does not claim clinical deployment or that a correction for the observed N3 failure mode has been validated.

The manuscript is original, has not been published previously, and is not under consideration by another journal. Both authors have approved the manuscript and agree with its submission. The authors declare no competing interests. Funding, ethics, data availability, CRediT authorship and use of generative AI are disclosed in the manuscript. Sleep-EDF is publicly available through PhysioNet; access to de-identified SHHS data is managed by the National Sleep Research Resource, and restricted participant-level data are not redistributed.

Thank you for considering our work.

Sincerely,

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